

## Mid-term Exam (Graph Neural Networks - Fall 2026) [예상 문제지]

This paper follows the structure the instructor announced on 2026-10-06:

- Exam on the Tuesday of week 8 (October 20, 2026)
- Coverage: weeks 2 to 7. Week 1 is excluded.
- About 12 questions, roughly two per week.
- Week 2 has only the message passing mechanism, so it may carry one combined question instead of two.

Questions 1 to 9 below follow that order:

- 1      week 2    GCN
- 2, 3   week 3    1-WL colour refinement, GIN
- 4, 5   week 4    GraphSAGE, Cluster-GCN
- 6, 7   week 5    GCNII, PairNorm
- 8, 9   week 6    Homophily metrics, Label Informativeness

Week 7 material has not been released yet, so the two week 7 questions are left out. They will be added once the slides are published.

Full solutions for every question are on the study site under the same model name.

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Note: Students should write in English for evaluation.

**1. (GCN)** Given a graph with an adjacency matrix  $A$  and initial node feature matrix  $H^{(0)}$  as follows:

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} 1 \\ -1 \\ 2 \\ 0 \\ 3 \end{bmatrix}$$

Assume that the hidden layer of a GCN model of all nodes at layer  $(k)$  can be calculated as:

$$H^{(k)} = \sigma(\tilde{A} \cdot H^{(k-1)}),$$

where  $H^{(k)}$  denotes the output at layer  $k$ ,  $\tilde{A}$  is the normalized matrix ( $\tilde{A} = D^{-1}A$ ), and  $\sigma$  is a ReLU function  $\text{ReLU}(x) = \max(0, x)$ .

- a) Explain how information is propagated in a GCN layer.
- b) Calculate  $\tilde{A}$ .
- c) Calculate the output of the GCN model at layer  $k = 1$ .

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**2. (1-WL Test)** Consider two undirected graphs  $G_1$  and  $G_2$  with five nodes each, given in the following figures.

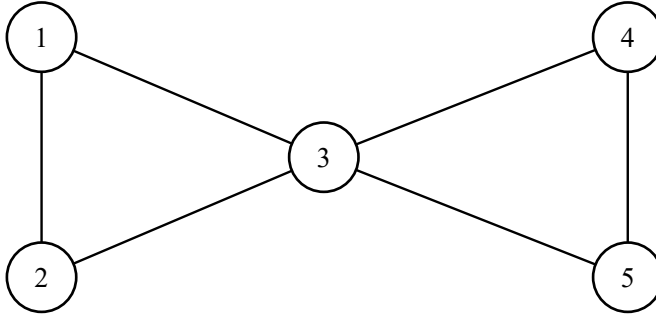


Figure 1. Graph  $G_1$

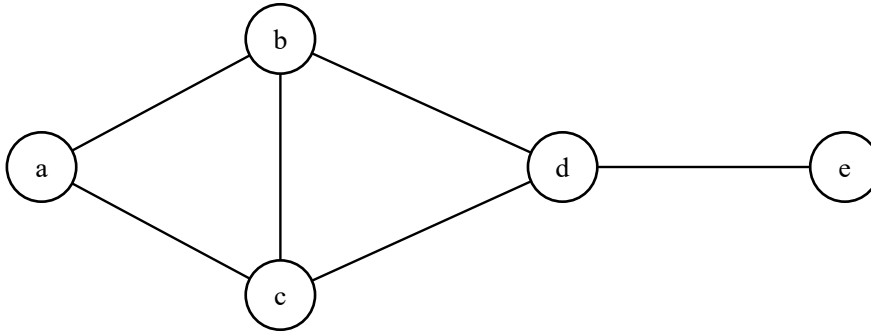


Figure 2. Graph  $G_2$

The 1-WL (Weisfeiler-Lehman) color refinement algorithm assigns every node the same initial color  $c^{(0)}(v) = 1$ , and then repeats:

$$c^{(k+1)}(v) = \text{HASH}_{\text{left}}(c^{(k)}(v), \{c^{(k)}(u) \}_{u \in N(v)} \text{right})$$

where HASH gives a new color to each distinct pair, and the neighbor colors form a multiset.

- a) Explain the 1-WL color refinement algorithm and what its result can and cannot prove.
- b) Run two rounds of color refinement on  $G_1$  and report the color classes.
- c) Run the same on  $G_2$  and decide whether  $G_1$  and  $G_2$  can be isomorphic.

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**3. (GIN)** Consider an undirected graph  $G$  with an adjacency matrix  $A$  and an initial node feature matrix  $H^{(0)}$  of one-hot colors as follows:

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

A Graph Isomorphism Network (GIN) layer is defined as:

$$h_v^{(k)} = \text{MLP}\left((1 + \varepsilon) \cdot h_v^{(k-1)} + \sum_{u \in N(v)} h_u^{(k-1)}\right)$$

where  $\varepsilon = 0.1$ . Assume the MLP is a single linear layer followed by ReLU,  $\text{MLP}(x) = \sigma(xW)$ , with

$$W = \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix}$$

and  $\sigma$  is a ReLU function  $\text{ReLU}(x) = \max(0, x)$ .

- a) Explain why GIN is the most expressive message passing GNN.
- b) Calculate the aggregated values inside the MLP for all nodes at layer  $k = 1$ .
- c) Calculate  $H^{(1)}$ , and compare the aggregation with that of a standard GCN layer.

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**4. (GraphSAGE)** Consider an undirected graph  $G$  of five nodes A, B, C, D and E given in the following figure. Each node has initial features that are the numbers standing next to it (i.e. the initial feature of node 'A' is  $h_A^{(0)} = 0.4$ ). According to the GraphSAGE model with an AGGREGATE that is a MEAN function, the feature of a node  $i$  at layer  $k$  can be updated as:

$$h_{N(i)}^{(k)} = \text{AGGREGATE}(\{ h_u^{(k-1)}, \forall u \in N(i) \})$$

$$h_i^{(k)} = \text{ReLU}( h_i^{(k-1)} \parallel h_{N(i)}^{(k)} )$$

where  $\parallel$  is a concatenation,  $\text{ReLU}(x) = \max(0, x)$ , and  $N(i)$  is the set of neighbour nodes of node  $i$ .

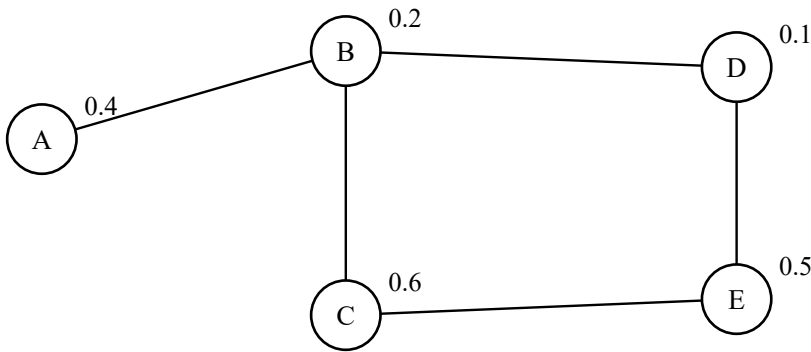


Figure 2. Graph G

- a) What is the main idea behind the GraphSAGE model, and how does it differ from traditional GCNs?
- b) Calculate the feature of each node at  $k = 1$ .
- c) Calculate a graph-level embedding  $h_G$  by using a 'Mean' global pooling when  $k = 1$ .

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**5. (ClusterGCN)** Consider an undirected graph  $G$  of six nodes A, B, C, D, E and F given in the following figure. The graph  $G$  contains two clusters  $G_1$  and  $G_2$ . Each node has initial features that are the numbers standing next to it. According to the ClusterGCN model, the feature of a node  $i$  at layer  $k$  can be updated as:

$$h_{N(i)}^{(k)} = \text{MEAN}(\{ h_u^{(k-1)}, \forall u \in N(i), G_u = G_i \})$$

$$h_i^{(k)} = \text{ReLU}( h_i^{(k-1)} \parallel h_{N(i)}^{(k)} )$$

where  $\parallel$  is a concatenation and  $\text{ReLU}(x) = \max(0, x)$ .

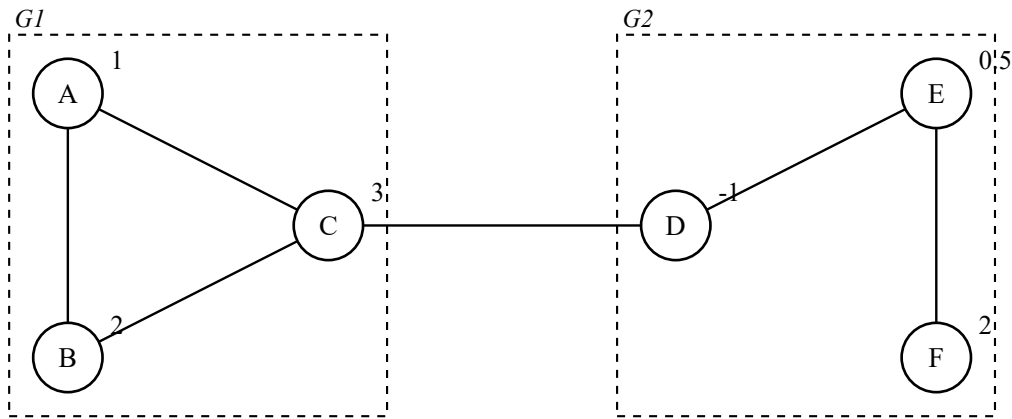


Figure 3. Graph G with two clusters

- a) What is mini-batch training, and why does ClusterGCN need it?
- b) Calculate the output representations of all nodes at layer  $k = 1$ .

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**6. (GCNII)** Given a graph with an adjacency matrix  $A$  and initial node feature matrix  $H^{(0)}$  as follows:

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix}$$

$$H^{(0)} = \begin{bmatrix} 3 \\ -1 \\ 1 \\ 2 \\ -3 \end{bmatrix}$$

Assume that the output of a GCNII model of all nodes at layer  $(k)$  can be calculated as:

$$H^{(k)} = \sigma[ ((1 - \beta) I_n) \cdot ((1 - \alpha) \tilde{A} \cdot H^{(k-1)} + \alpha H^{(0)}) ]$$

where  $H^{(k)}$  denotes the output at layer  $k$ ,  $\tilde{A}$  is the normalized matrix ( $\tilde{A} = D^{-1}A$ ),  $I_n$  is the identity matrix,  $\alpha = \beta = 0.5$ , and  $\sigma$  is a ReLU function  $\text{ReLU}(x) = \max(0, x)$ .

- a) What are the two major components introduced in GCNII to overcome the depth limitation problem in GCNs?
- b) Calculate  $\tilde{A}$ .
- c) Calculate the output representations at layer  $k = 1$ .

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**7. (PairNorm)** After one GCN layer the node features of a four-node graph are

$$X = \begin{bmatrix} 3 & 1 \\ 1 & -1 \\ -1 & 2 \\ 1 & 2 \end{bmatrix}$$

PairNorm normalizes them in two steps, a centering step and a scaling step:

$$x_i^c = x_i - (1/n) \sum_{i=1}^n x_i \quad (\text{Center})$$

$$x_i = s \cdot x_i^c / \sqrt{(1/n) \sum_{i=1}^n \|x_i^c\|_2^2} \quad (\text{Scale})$$

with  $s = 1$  and  $n = 4$ .

- a) Explain node-wise and feature-wise over-smoothing, and explain which quantity PairNorm keeps constant across layers.
- b) Calculate the centered features  $x_i^c$ .
- c) Calculate the scaling denominator and the final  $x_i$ .

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**8. (Homophily Metrics)** Consider the undirected graph  $G$  below. The number next to each node is its class label  $y$ .

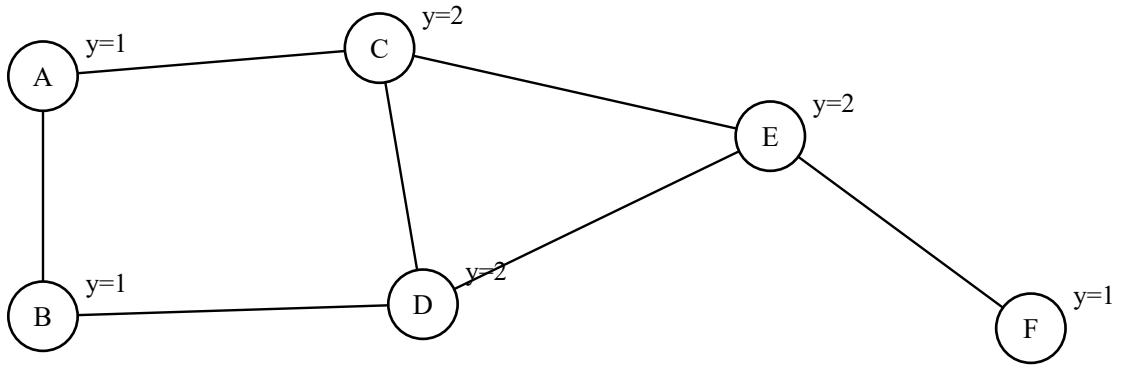


Figure 1. A graph with two classes

Four homophily metrics are defined as follows ( $C$  is the number of classes):

$$h_{edge} = |\{(u,v) \text{ in } E : y_u = y_v\}| / |E|$$

$$h_{node} = (1/|V|) \sum_v |\{u \text{ in } N(v) : y_u = y_v\}| / d(v)$$

$$h_{class} = (1/C) \sum_k \max(0, h_k - |V_k|/|V|), \quad h_k = |\{(u,v) \text{ in } E : y_u = y_v = k\}| / |\{(u,v) \text{ in } E : y_u = k\}|$$

$$h_{adj} = (h_{edge} - \sum_k p_k^2) / (1 - \sum_k p_k^2), \quad p_k = \sum_{v \text{ in } V_k} d(v) / 2|E|$$

- Explain why edge homophily can be misleading, and explain what node homophily and adjusted homophily each correct.
- Calculate  $h_{edge}$  and  $h_{node}$ .
- Calculate  $h_{class}$  and  $h_{adj}$ , and state whether this graph is homophilic.

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**9. (Label Informativeness)** Consider the two graphs below. Both have four nodes ( $y_A = y_B = 1, y_C = y_D = 2$ ) and four edges.



Figure 1. Graph G1 (left: A, B, C, D) and Graph G2 (right: E, F, G, H)

Label informativeness measures the mutual information between the labels at the two ends of an edge:

$$LI = I(Y_u ; Y_v) / H(Y_u)$$

$$I(Y_u; Y_v) = \sum_{p,q} P(p,q) \log_2 ( P(p,q) / (P(p) P(q)) ), \quad H(Y_u) = - \sum_p P(p) \log_2 P(p)$$

$P(p,q)$  is the probability that a randomly chosen directed edge goes from a node of class  $p$  to a node of class  $q$ .

- Explain why edge homophily alone can be a misleading measure of heterophily.
- Calculate  $h_{edge}$  and  $LI$  for graph G1.
- Calculate  $h_{edge}$  and  $LI$  for graph G2, and explain which graph a GCN can solve.